MIT NSE | 22.S906 Fusion Engineering
Neutronics Project

Due date: TBD
Revision: 07 Oct 26

Objective

Monte Carlo particle transport codes are the principal nuclear design and analysis tool for fusion energy systems. MC codes such as OpenMC or MCNP are employed throughout the lifetime of a fusion energy system, from simple but fast models for rapid iteration in the design phase to massive, high fidelity, and CPU intensitve models for operations and safety analysis on an as-built machine. The combination of modern open-source MC codes like OpenMC combined with generative AI should enable students with little background experience to start using these tools to do basic design and analysis.

The objectives of this assignment are:

1. Understand the basic inputs and outputs of an MC model for a fusion power plant
2. Use the model to analyze the tradeoffs imposed by the nuclear aspects of an FPP
3. Practice using generative AI to enable more complex model but also practice reviewing, modifying/commenting, and understanding the code it generates so you remain in control
4. Develop a platform that you can use to do nuclear analysis of fusion power plant subsystems that we will encounter later in the course and/or in your future professional work after this course.

Preparation

The assignment is composed of the follow parts:

1. Download and install OpenMC: OpenMC is an open-source, modern, powerful, and widely-used neutronics code for fission and fusion particle transport modeling. It originated as the NSE Ph.D. project of Paul Romano back in the late 2000s but is now under development by world-wide network of collaborators. You should down and install the code on your computer. There are many source of assistance, including AI tools, your fellow classmates in NSE who use OpenMC, and your instructor. Please work hard on your own to do this, but reach out if you have difficulties:
   https://openmc.org/
   https://docs.openmc.org/en/stable/usersguide/install.html#usersguide-install

2. Download nuclear data libraries: An essential input to OpenMC (or any other Monte Carlo particle transport code) is the nuclear data that will be used to model how particles move through matter. Please download and install the following nuclear data libraries for OpenMC: (1) ENDF/B-VIII.1 and (2) FENDL 3.2. You can download thse libraries direcly from the OpenMC webpage:
   https://openmc.org/data/

3. Verify your installation and data: You should run one of the examples that is included with the OpenMC installation to ensure your code is working properly. You can also find a lot of examples in convenient Jupyter Notebook format here: https://docs.openmc.org/en/v0.12.2/examples/index.html

Assignment

In short, the first part of the assignment is to build a generic, parameterized tokamak OpenMC model of a fusion power plant and verify that it works. Then, as the semester progress, I will add additional parts to this document in which you will use your OpenMC model to perform some relevant nuclear analysis for the sub-system that we are discussing.

1. Build your own "Generomak" OpenMC Model: You and/or your favorite generative AI tools should develop a basic OpenMC model of a tokamak-based fusion power plant. The model can be as simple or as complex as you like (there are tradeoffs either way). You decide how advanced of a neutron source you want. You decide if you want to do a simple 1D radial build model with slabs, a "1.5D" model of nested toroids, or more complex 2D or 3D geometry.
    a. Source: You should implement a realistic source of neutrons that is representative (in geometry, energy, angle, etc.) of a thermonuclear D-T plasma. Ensure you do some research to understand the relevant properties.
    b. Geometry: You should (at minimum) have the following major subsystems of the fusion power plant along with the materials that compose them:
        i. Plasma
        ii. First wall (divertor optional depending on your geometry complexity)
        iii. Vacuum vessel (ports optional depending on your geometry complexity)
        iv. Blanket
        v. TF magnet
        vi. CS magnet
        vii. PF/EF magnets
    c. Tallies: You should implement a single tally at the present time: The average neutron scalar flux spectrum within the first wall.

    You are welcome and encouraged to use generative AI to assist in building your Generomak model, particularly if you'd like to try more complex modeling of the source and geometry; however, you are strongly recommended to review and thoroughly understand all of your input source code so that ensure knowledge and oversight of your model.